

Discussion of Bridging Structured Knowledge and Data

Structured-Knowledge-Informed Neural Networks (SKINNs)

[Discussant]

The paper in one slide

A flexible black box $f_\theta(X)$ and a differentiable “theory” $g_\phi(X^{\text{SK}})$, trained *together*.

$$\mathcal{L}(\theta, \phi) = \underbrace{\frac{1}{N} \sum_i (f_\theta(X_i) - y_i)^2}_{\text{fit the data}} + \lambda \underbrace{\frac{1}{M} \sum_j (f_\theta(X_j^{\text{grid}}) - g_\phi(X_j^{\text{SK}}))^2}_{\text{stay near theory on a grid}}.$$

- $\phi = \text{learnable}$ latent economic states (volatility, a risk-neutral density).
- Three flavors of g_ϕ : **parametric** (BS, Heston, jumps), **semi-parametric** (a frozen deep surrogate of an expensive simulator), **non-parametric** (a 2,000-dim risk-neutral density on a simplex, or an autoencoder).

Two claims

Theory: SKINNs are M -estimators $\Rightarrow \sqrt{N}$ -consistent, asymptotically normal, sandwich covariance. **Empirics:** on S&P 500 *vanilla calls* (1996–2022) they beat NN, PINN, transfer learning, and structural models.

Four lenses, one object

The composite loss is **regularized empirical risk minimization** with a moving target. Everything else is a re-reading of that single idea:

- **GMM**: two moments $m_1 = f_\theta - y$, $m_2 = f_\theta - g_\phi$, weight $W = \text{diag}(1, \lambda)$, deliberately *sub-optimal* $\Rightarrow$ regularization, not efficiency.
- **Bayesian MAP**: Gaussian likelihood + a Gaussian “theory prior” centered at g_ϕ , precision λ ; the prior center is *learned* $\Rightarrow$ empirical Bayes.
- **Variational / physics**: minimize an energy; but the constraint is the *solution* g_ϕ , not a differential operator.
- **M-estimation**: $\hat{\beta} = \arg \min_{\beta} \frac{1}{N} \sum \rho(\cdot; \beta)$, $\beta = (\theta, \phi)$.

The framing is honest and well-curated. My questions are about *which* claims the framing actually licenses.

Elegant idea, the right diagnosis.

Two points I'll press: the asymptotic statements can be sharpened, and the study focuses on the most favorable corner of the range.

Two real strengths → six points → a constructive "SKINN 2.0".

Strength I: the deep one, constrain the solution, not the operator

Why do the gradient histograms (their Fig. 6 vs. 7) look so different?

- A **PINN** penalizes a differential operator, $\|\partial_t f + \mathcal{N}[f]\|^2$. Differentiating the network multiplies its *neural tangent kernel* (how a wide net learns) eigenvalues by powers of frequency: high modes are amplified, the kernel becomes ill-conditioned, and gradients across loss terms decorrelate, the known PINN gradient difficulty.
- **SKINN** penalizes a *zeroth-order* quantity $f_\theta - g_\phi$. The NTK is left essentially intact, so the three loss terms share curvature and train together.

Plainly

PINNs ask the net to satisfy an equation by differentiating itself; SKINNs ask it to match an *answer*. The second is numerically far kinder, and it also handles *unobservable* state variables (Heston's instantaneous variance) that are out of reach for PINNs. This is, in my view, the paper's most valuable and least-advertised contribution.

Strength II: joint learning as a learned-center prior

Calibration and two-stage transfer learning fix the theory first, then fit. SKINN lets θ and ϕ *co-evolve*: the prior's center g_ϕ moves toward the data while the net is shrunk toward the prior.

- This is empirical Bayes done by backprop, and it is the right instinct.
- It also rescues the surrogate route: pre-train a network to emulate an expensive simulator, freeze it, learn ϕ through it. Surrogate emulation of expensive simulators is an idea with independent traction in the pricing literature, so the engineering instinct is well supported.

Point 1: whose $\sqrt{N}$ is it?

Theorems 1–2 assume a *unique interior minimizer* and an *invertible Hessian*. For an over-parameterized network these do not hold:

θ has permutation and ReLU-rescaling symmetry $\Rightarrow \nabla^2 \mathcal{L}$ is singular, θ^* not unique.

The fix is known, and it strengthens the paper

Do **not** claim normality for the weights θ . Treat f_θ as an infinite-dimensional *nuisance*; profile it out, $\theta(\phi)$, and build a **Neyman-orthogonal** score (so small errors in the flexible part don't bias ϕ) for the low-dimensional ϕ :

$$\partial_\theta \psi(\phi, \theta(\phi)) = 0 \quad (\text{no first-order leakage from the nuisance}).$$

Then $\sqrt{N}(\hat{\phi} - \phi^*) \rightarrow \mathcal{N}$ survives: this is the double/debiased-ML route, and it is the *correct* vehicle for “inference on interpretable parameters.”

Point 2: is ϕ identified, and relative to what?

- **Relevance.** ϕ is identified only if g_ϕ genuinely bends with it on the grid:

$$\text{rank} \left(\mathbb{E}_\nu [\partial_\phi g_\phi(X^{\text{SK}})] \right) = \dim \phi.$$

A near-flat g_ϕ in ϕ over the traded region is the option-pricing analogue of a *weak instrument*; Wald intervals then under-cover. Use identification-robust inference.

- **Grid-dependent estimand.** $\mathcal{L}_{\text{SK}}$ integrates over a *collocation measure* ν (the grid you score theory on) broader than the data law. Hence what ϕ estimates depends on the grid:

$$\phi^* = \phi^*(\nu, \lambda) \quad \text{is the analyst's object, not a market primitive.}$$

The economic reading of “the implied volatility” inherits this dependence. **Remedy:** declare ν , report sensitivity, weight ν toward liquidity.

Point 3: no-arbitrage lives in g_ϕ , but you price with f_θ

The constrained object is g_ϕ ; the reported surface is f_θ , tied to it only by a *soft* λ , so no-arbitrage is enforced softly. Three consequences:

- Reported SKINN prices need not be arbitrage-free at all, the guarantee is one λ -slack away.
- **MOPA** pins only the per-maturity *marginal* (Breedon–Litzenberger). The joint law is free $\Rightarrow$ forward-starting / path-dependent payoffs are *unpriced*.
- A piecewise-constant-in- τ density across the 10 fixed tenors may not satisfy **convex order** (later densities must be more spread) $\Rightarrow$ calendar arbitrage.

Remedy

Price from g_ϕ (or hard-project f_θ onto the no-arbitrage cone), and impose convex order across maturities. Arbitrage-freeness must be *structural*, not a penalty.

Point 4: the perimeter, and what the frontier already knows

The test bed is *European S&P 500 vanilla calls*, one index, calls only, 7–365 days.

- No **arbitrage-free surface** baseline (SVI/SSVI/GP), the real desk standard. Beating plain-NN or one- σ Black–Scholes is a baseline worth strengthening.
- RMSE *in price* is dominated by intrinsic value; the informative part (implied vol) is a sliver. Report IV- and vega-weighted error.
- Worth citing the close prior work: inverse PINNs (already co-estimate parameters), differentiable calibration / neural-SDE, no-arbitrage neural SDF, and risk-neutral density estimation (Breedon–Litzenberger / Jackwerth / Fengler).

Guyon's finding, made concrete

SPX prices carry *no* forward-volatility information (Bourgey–Guyon). And when a neural object *is* fit under the right hard constraints (Guyon–Mustapha, neural joint SPX/VIX), it learns the vol factor is **path-dependent and mean-reverting**, $\rho = -1$. The “interpretable Markovian state” SKINN hard-codes is a structure recent evidence points away from.

Point 5: pricing-optimal $\neq$ hedging-optimal

SKINN's Delta is autodiff of a *price*-fitted net. Fitting a level does not control its slope: AD-Greeks of an MSE-trained network are typically imprecise. Their own observation, “boundary conditions don't improve hedging,” is exactly this.

Hedging needs $\|\partial_m f_\theta - \partial_m g_\phi\|^2$ in the loss: **Sobolev training**.

Plainly

If you want good Greeks, *penalize the Greeks*, or make the hedging P&L the primary objective (deep hedging). This single change would tie together why their derivative-regularized net helps while their boundary net does not improve hedging.

(Also: their Delta from degree-1 homogeneity should be two terms, $\Delta = c(m) - \frac{K}{S} c'(m)$ with $C = S c(m)$, not one.)

Point 6: the evaluation

- Several comparisons are **nested** (more knowledge $\supset$ less). Plain Diebold–Mariano is invalid there, so use **Clark–West**.
- Hundreds of pairwise tests, no correction $\Rightarrow$ inflated stars. Use a **Model Confidence Set** or Romano–Wolf.
- The “ ϕ predicts next-month VIX” regressions mix $\mathbb{P}$ (realized VIX) with $\mathbb{Q}$ (pricing parameters). A $\mathbb{P}$ -forecast does not validate a $\mathbb{Q}$ -price.
- Hedging P&L has **no transaction costs** and a single one-day horizon.

Smaller fixes (for the authors)

- AE map written $f^{\text{enc}} \circ f^{\text{dec}}$, order reversed; should be $f^{\text{dec}} \circ f^{\text{enc}}$.
- θ reused for both NN weights *and* a Heston parameter in the appendix.
- Sampling: “uniform” in §3.2 vs. “LHS/grid” in the appendix, which?
- Table 3 panels duplicate columns; a statistic (-12.95) recurs across horizons.
- Figs. 11–12 share a caption; one is the shorter horizon.
- Hedging caption says “squared” but the metric is mean *absolute* error.
- Broken cross-reference “Appendix ??”.

None fatal, but a 9-dim surrogate trained on 5×10^7 points is ≈ 6 points per axis; report held-out surrogate error.

A constructive “SKINN 2.0”

Keep the framework; change the instantiation so it enters the arena the field cares about:

1. g_ϕ = **path-dependent volatility** (Guyon–Lekeufack), not Markovian Heston.
2. **Add VIX** futures and VIX-smile moments, with the *log-contract identity* enforced as a **hard** constraint (couples the two markets, no-arbitrage by construction).
3. **Sobolev / deep-hedging loss** so Greeks and hedging P&L are first-class.
4. **Orthogonalized inference** for ϕ ; declare the collocation measure ν .
5. **Benchmark** against an arbitrage-free surface; evaluate with an MCS, IV-space errors, and net-of-cost hedging.

The idea is elegant and the diagnosis is right.

As stated it is a statistical estimation framework, not yet a pricing engine; a few additions would make it a competitive one.

The test bed is favorable to it, and the fixes are clear, principled, and within reach.